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TECTONIC INTERPRETATION OF PALEOMAGNETIC RESULTS FROM THE UPPER DEVONIAN GRIOTTE OF THE MONTAGNE NOIRE, SOUTHERN FRANCE

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ABSTRACT

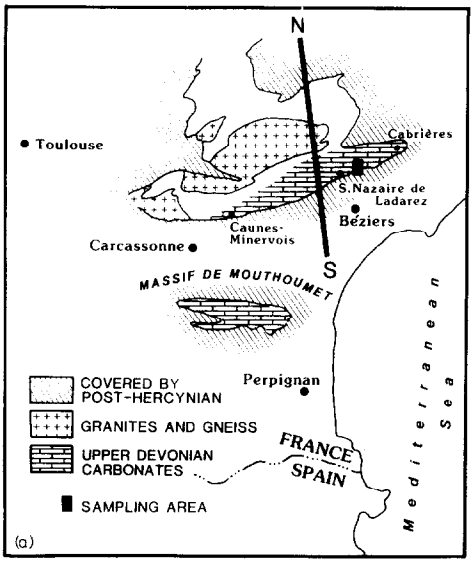
Coudray, J., Edinger, P.F., Michel, D. and Nairn, A.E.M., 1986. Tectonic interpretation of paleomagnetic results from the Upper Devonian griotte of the Montagne Noire, southern France. *Tectonophysics*, 126: 137–151.

The Devonian (Famennian) griotte acquired its magnetization during and subsequent to emplacement during the Late Carboniferous–Early Permian. The absence of normal secondary components suggests that tectonic emplacement postdates the period of frequent Carboniferous reversals, that is post-Westphalian A and presumably after the onset of the Kiamian quiet period. A remanence attributed to the Permian, as found in the south of France, has been isolated in a few samples; the remainder of the samples show a stable magnetization with a great circle distribution which passes through the zone of Permian directions.

INTRODUCTION

The Upper Devonian (Famennian) griotte, sometimes informally called the "goniatitico rosso," is a condensed red nodular limestone which shows many of the lithologic characteristics of the better known Jurassic "ammonitico rosso." It occurs in the Paleozoic succession of the Montagne Noire, where it crops out in the Mt. Peyroux nappe. The outcrop was sampled in an abandoned quarry near the village of St. Nazaire de Ladarez (Fig. 1). The beds stand vertically here, which facilitated the collection of a detailed section (Figs. 2 and 3). An objective of this study was to see if a magnetostratigraphy could be established. The sedimentologic model of

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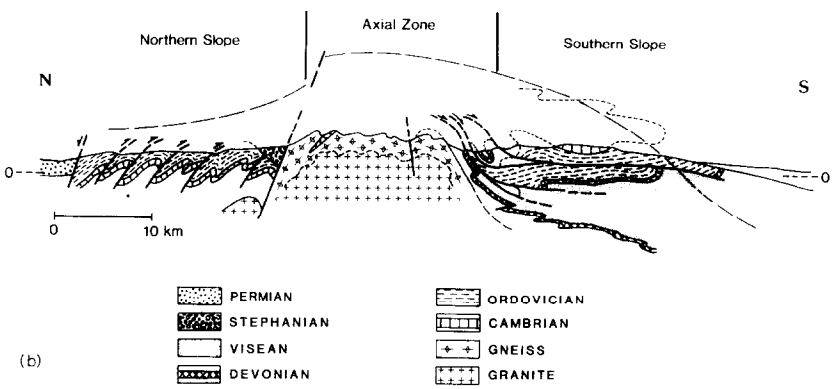


Fig. 1. a. Location map of the Paleozoic outcrops of the Montagne Noire. The area where the griotte was sampled near St. Nazaire de Ladarez is indicated by a solid block. b. Generalized cross section of the Montagne Noire showing the Mt. Peyroux nappe south of the metamorphic core of the Montagne Noire (from Michel, 1981).

Coudray and Michel suggested that the griotte formed on a slope and that nodularity was due to the breakup of an alteration of thin calcareous and marly beds by downslope movement (Coudray and Michel, 1980; Michel, 1981). There was thus the possibility that a marly matrix flow around nodules might be detected

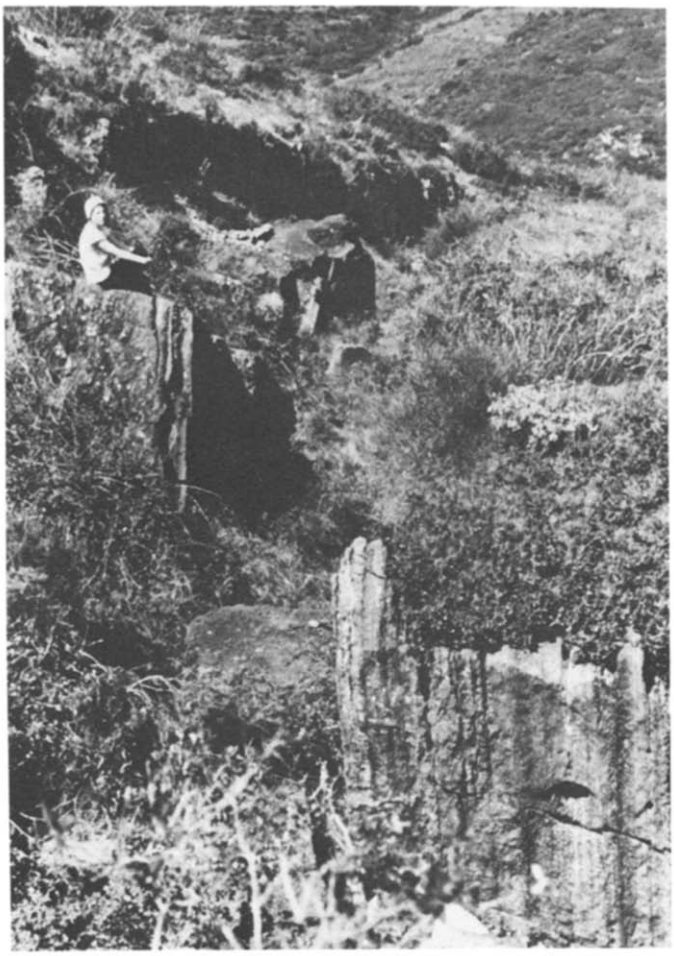


Fig. 2. General view of the old "Concours" quarry near St. Nazaire de Ladarez (photo by D. Michel).

if magnetization were primary. In fact, neither of these objectives was achieved and we concur with the subsequently published results and discussion of Rochette et al. (1982) that the magnetization of the griotte is secondary.

OUTLINE OF GEOLOGY

The Montagne Noire, at the extreme south of the Massif Central, comprises a broad zone of Paleozoic outcrops extending approximately 120 km east–west on either side of, and inclusive of, an axis of metamorphic rocks (Fig. 1a). Devonian rocks are not known in outcrop north of the axis, but are well represented along the southern flank. The griotte occurs within the Upper Devonian (Famennian) succes-

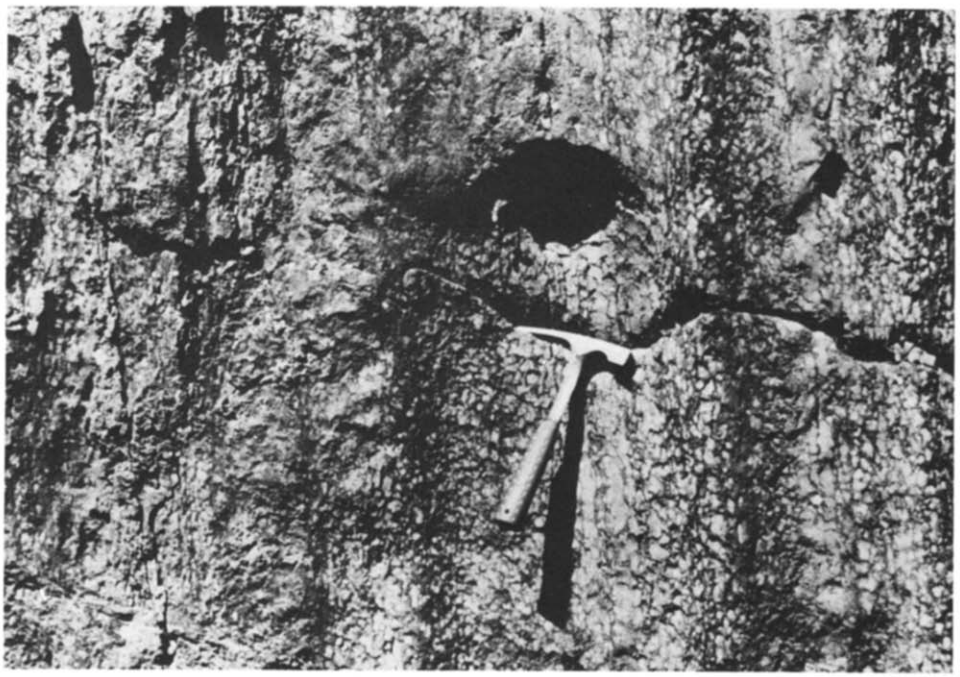


Fig. 3. Closeup of the griotte in the Concours quarry (photo by D. Michel).

sion. Boyer et al. (1968) retain the traditional description of the griotte, which dates back to Dufrenoy (1833; see Crilat, 1983), as a red nodular limestone with a certain proportion of clay rich in small ammonoids (2–3 cm). Sometimes the nodules may be enveloped by clay which may have a color other than red (green, in particular). The stratigraphy and biostratigraphy have been reviewed by Boyer et al. (1968) and Feist (1978), who have referred to the overlying and underlying sediments as supra- and infra-griotte, respectively, in terms of their relation to the griotte. The details of the section are summarized in Fig. 4 (from Michel, 1981). The detailed stratigraphic analysis (see Boyer et al. 1968) indicates that the sequence is condensed and that there is no break in sedimentation at the Devonian–Carboniferous boundary.

There has been much discussion of the mode of formation of these condensed, nodular, hematitic limestones. The arguments concerning the primary origin of the hematite have been reviewed by Nardi (1982). Jenkyns and Torrens (1971) regarded the ammonitico rosso as formed on seamounts, which adequately accounts for the condensed sequence. Michel (1981) and Coudray and Michel (1980) suggest that deposition occurred on a slope and that the nodular form was due to the breakup of carbonate bands as a result of downslope movement following slow tilting. Crilat (1983) provides a paleogeographic sketch illustrating the relationship of the Devonian facies around Mt. Peyroux.

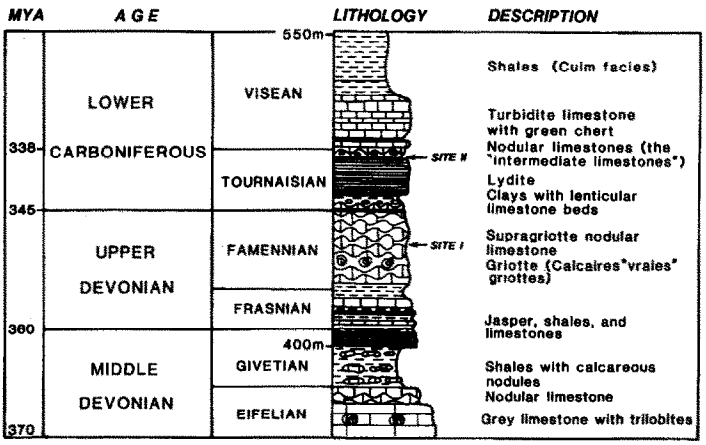


Fig. 4. The lithostratigraphic section of the Middle Devonian–Lower Carboniferous in the Mt. Peyroux nappe (after Michel, 1981).

The Paleozoic of the southern flank of the Montagne Noire was emplaced as a series of Variscan nappes derived from the north. The youngest rocks involved are Visean, although north of the metamorphic core of the Montagne Noire, Stephanian beds are involved in folding (Fig. 1b). A brief outline of structure is given by Engel et al. (1978).

PALEOMAGNETIC STUDY

A stratigraphic sequence of about 100 samples, spaced at approximately 10 cm intervals, was drilled from the vertically inclined griotte exposed in an old quarry denominated "Concours" which is on the southern slopes of the Montagne Noire. A second horizon of red nodular limestones, of Tournasian age, crops out on the hillside. Attempts to drill a stratigraphic sequence of these latter beds were frustrated, largely because of the closely spaced joints on the exposed weathered surfaces. Some cores were lost due to breakup during transit, others during slicing and thermal demagnetization.

Specimens were cut in the laboratory, and where cores yielded two specimens, these were used to provide the pilot samples for alternating field and thermal demagnetization utilizing Schonstedt equipment. The remanence was measured with a Digico magnetometer interfaced with a Digital MINC computer. The intensity of natural remanent magnetization ranged from as high as 4 A m^{-1} to 1 mA m^{-1}, with the average value near the lower end of the range (10–20 mA m^{-1}).

A preliminary suite of about 20 cores was selected for progressive demagnetization in alternating fields. The maximum field possible with the Schonstedt equipment (100 mT) was reached in nine steps. As the normalized demagnetization curves in Fig. 5a show, the response was varied, with M/M_0 0.1 to 0.6 of the original

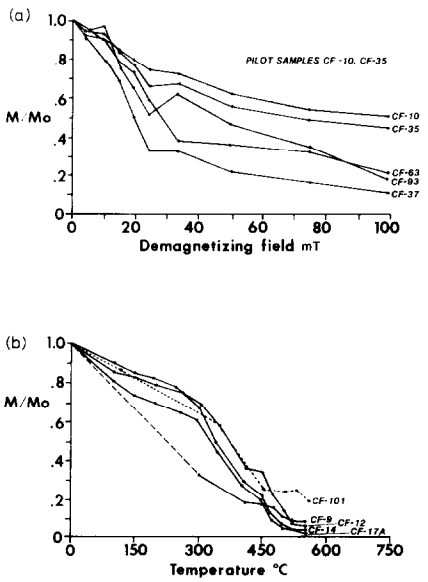


Fig. 5. a. Normalized alternating field demagnetization curves. b. Normalized thermal demagnetization curves.

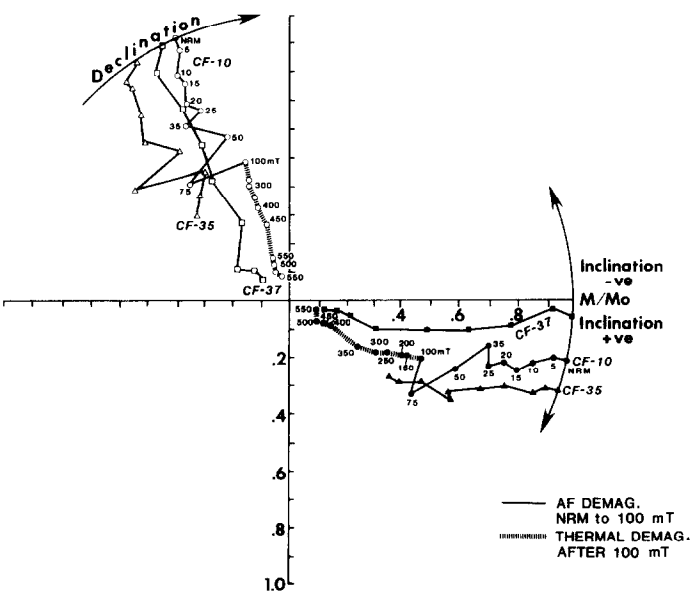


Fig. 6. Stable endpoint curves plotting normalized intensity against declination and inclination showing the response of griotte samples to alternating field demagnetization. Open symbols indicate declination; solid symbols indicate inclination. In the case of CF 10, the figures above 100 mT represent the results of thermal demagnetization following alternating field treatment.

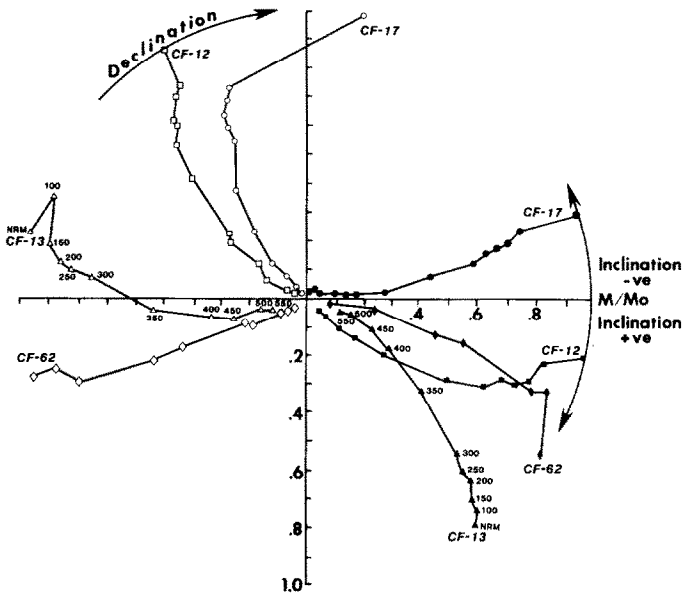
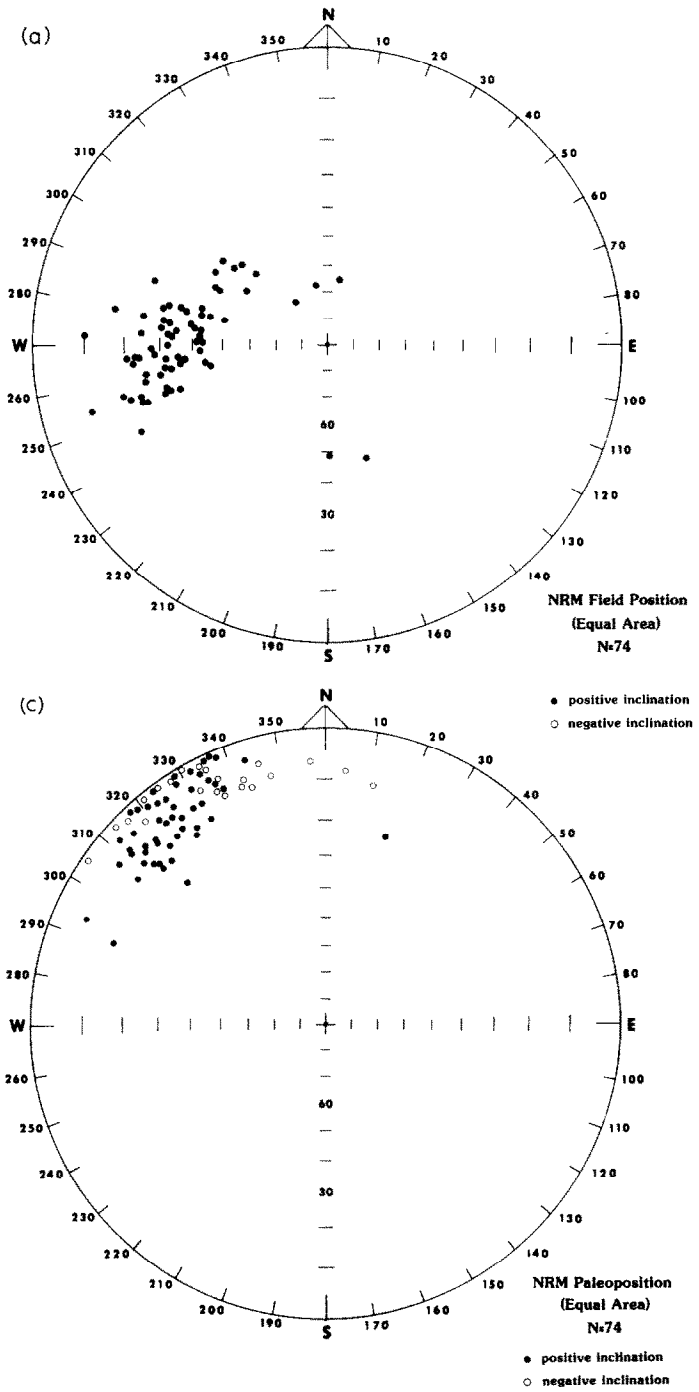


Fig. 7. Stable endpoint curves plotting normalized intensity against declination and inclination, showing the response of griotte samples to thermal demagnetization.

intensity remaining after demagnetization in fields of 100 mT. The stable endpoint curves (Fig. 6) also indicate that the characteristic magnetization had not been isolated.

Thermal demagnetization of pilot samples in the Schonstedt thermal demagnetizer proved to be a more successful technique (Fig. 5b), and as the stable endpoint curves of Fig. 7 show, a characteristic direction of remanence was generally isolated between 400°C and 450°C. The samples which had not responded to cleaning in alternating fields also yielded a characteristic remanence (Fig. 6). Consequently, all remaining samples were thermally demagnetized at 350°C, 400°C, and 450°C, and about half were taken to 500°C. The results were plotted as stereograms (Fig. 8) and in stratigraphic order (see Fig. 11). Fisher statistics of these 74 samples were computed and are shown in Table 1.

The directions of NRM and following thermal demagnetization at 450°C are shown in Fig. 8, with respect to both bedding and field position. A streaking of directions is apparent both before and after cleaning, and this can be shown even more clearly when the mean value is rotated to the center of the projection, following the technique of Creer et al. (1959), with all other directions moved by the same amount around small circles (see Fig. 9). This is not due to insufficient cleaning since the magnetic vector removed and the remaining vector are the same. A stereogram of these two vectors (Fig. 10) derived from straight line segments of the stable endpoint curves illustrate the point.


(a)
N
S
W
E
NRM Field Position
(Equal Area)
N=74
(c)
positive inclination
negative inclination
NRM Paleoposition
(Equal Area)
N=74
positive inclination
negative inclination

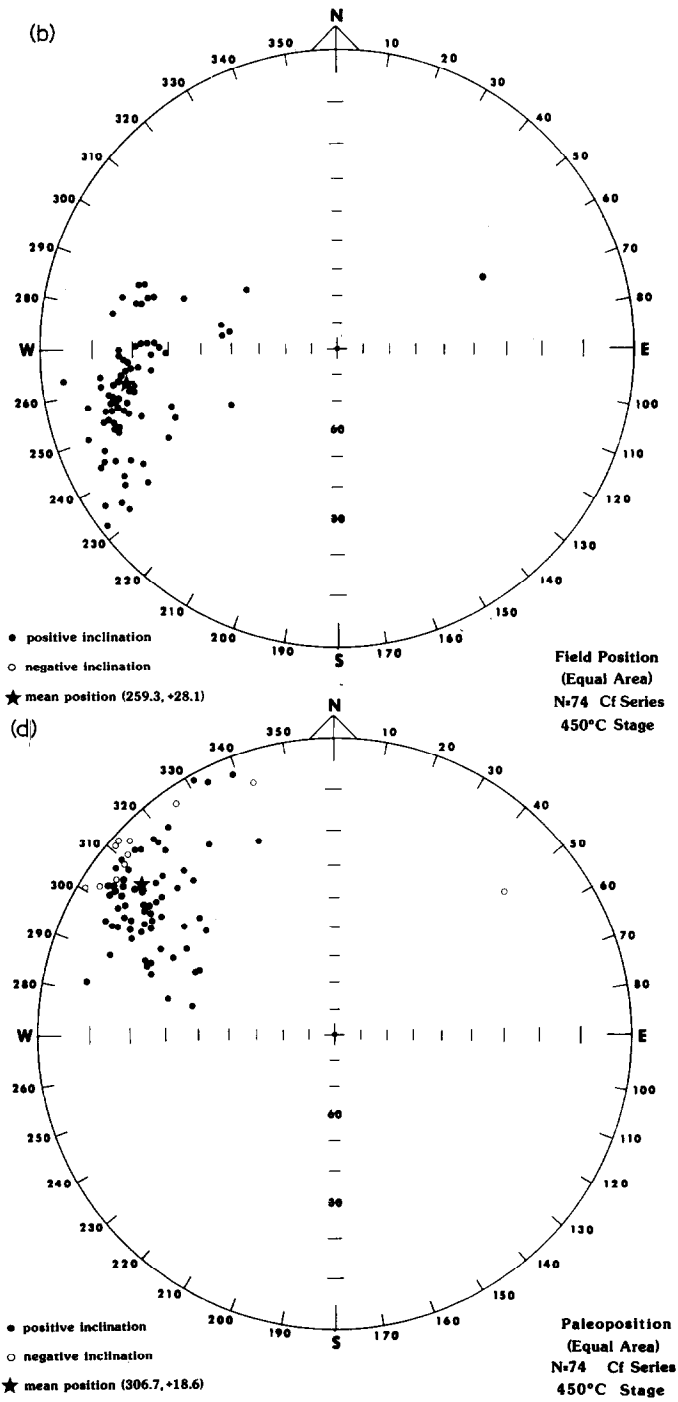


Fig. 8. Equal area plots of the NRM directions and direction of magnetization after thermal demagnetization at 450°C. Directions are plotted with respect to paleoposition (c, d) and to field position (a, b).

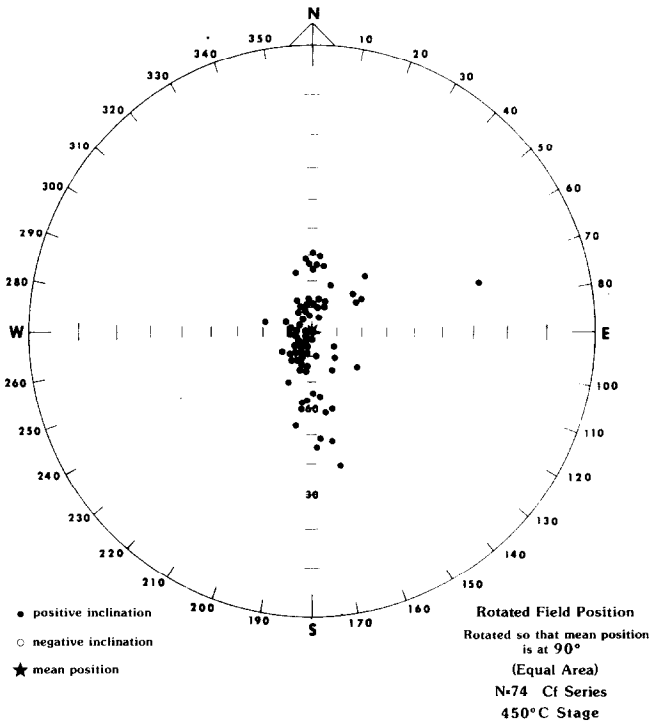


Fig. 9. The data of Fig. 8c with the mean direction rotated to the center of the projection and individual directions rotated along corresponding small circles to illustrate the streaking of the data.

TABLE 1

Summary of statistical parameters from the griotte samples

	Decl. (°)	Incl. (°)	N	α_{95} (°)	K	R	Temp. (°C)
Field position	259.3	+28.1	74	4.4	15	69.11	450
Paleoposition	306.7	+18.6	74				
Anomalous samples							
Field position	231.0	−15.3	7	19.4	11	6.44	450
Paleoposition	255.5	+46.1					
Sample CF 44							
Stable endpoint	183.4	−33.7					625
Vector removed	184.3	−36.9					575–625
Sample CF 46							
Hard component	213.2	−38.9					550
Soft component	238.2	−41.9					450–550
Zijderveld Permian	207.0	−22.0	Maures-Esterel Massif				
Kruseman Permian	198.5	−6.5					

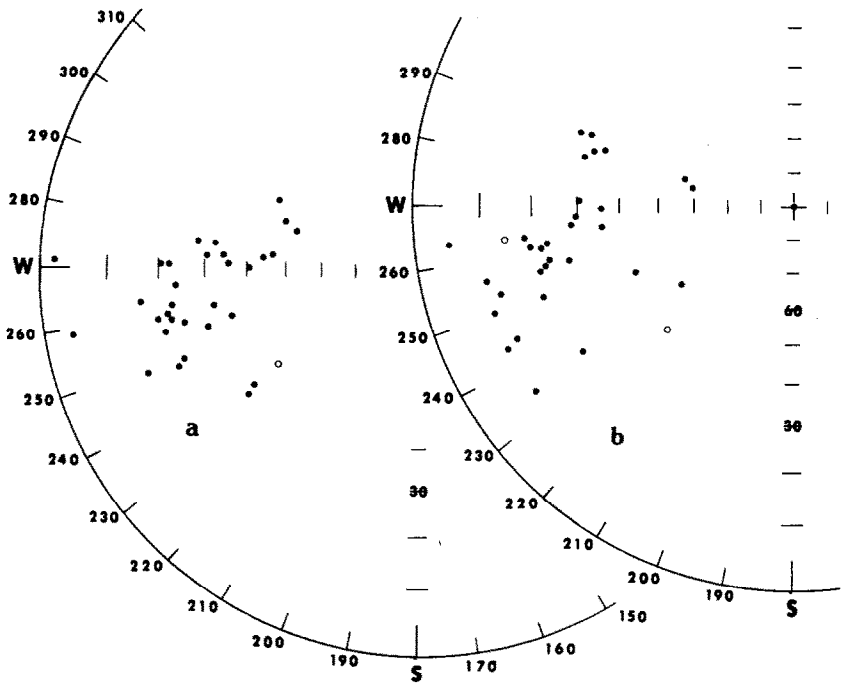


Fig. 10. Stereograms: (a) of the direction of magnetization after thermal demagnetization at 450°C; and (b) of the vector removed over the range 300°–500°C from straight line segments of the stable end point curves of the same samples.

To examine whether the streaking was systematic, the direction of magnetization was plotted in stratigraphic order (Fig. 11). As is clearly apparent in the smoothed curves (Fig. 11b), the variation appears to be more cyclical than systematic, with a tendency for the higher inclination values to be associated with low (270°–285°) declinations. On the declination curve of Fig. 11a, which shows plots for individual samples, the declination of nine samples deviated by more than 30° from the mean; these examples were re-examined. Two of them were discarded because of inconsistent results; in five cases (Fig. 12), a stable endpoint was not reached.

In only two of the anomalous samples characterized by negative inclination values was it possible to identify a stable endpoint at higher temperatures (Table 1) in which both the residual vector and the vector removed have the same directions. These directions approach the Permian direction of magnetization recorded by Zijderveld (1975) and Kruseman (1962) from the Maures-Esterel Massif. In the remaining five cases, although the residual moment had a negative inclination, the stable endpoint was not reached and the vector removed resembled the streaked directions of the majority of the samples.

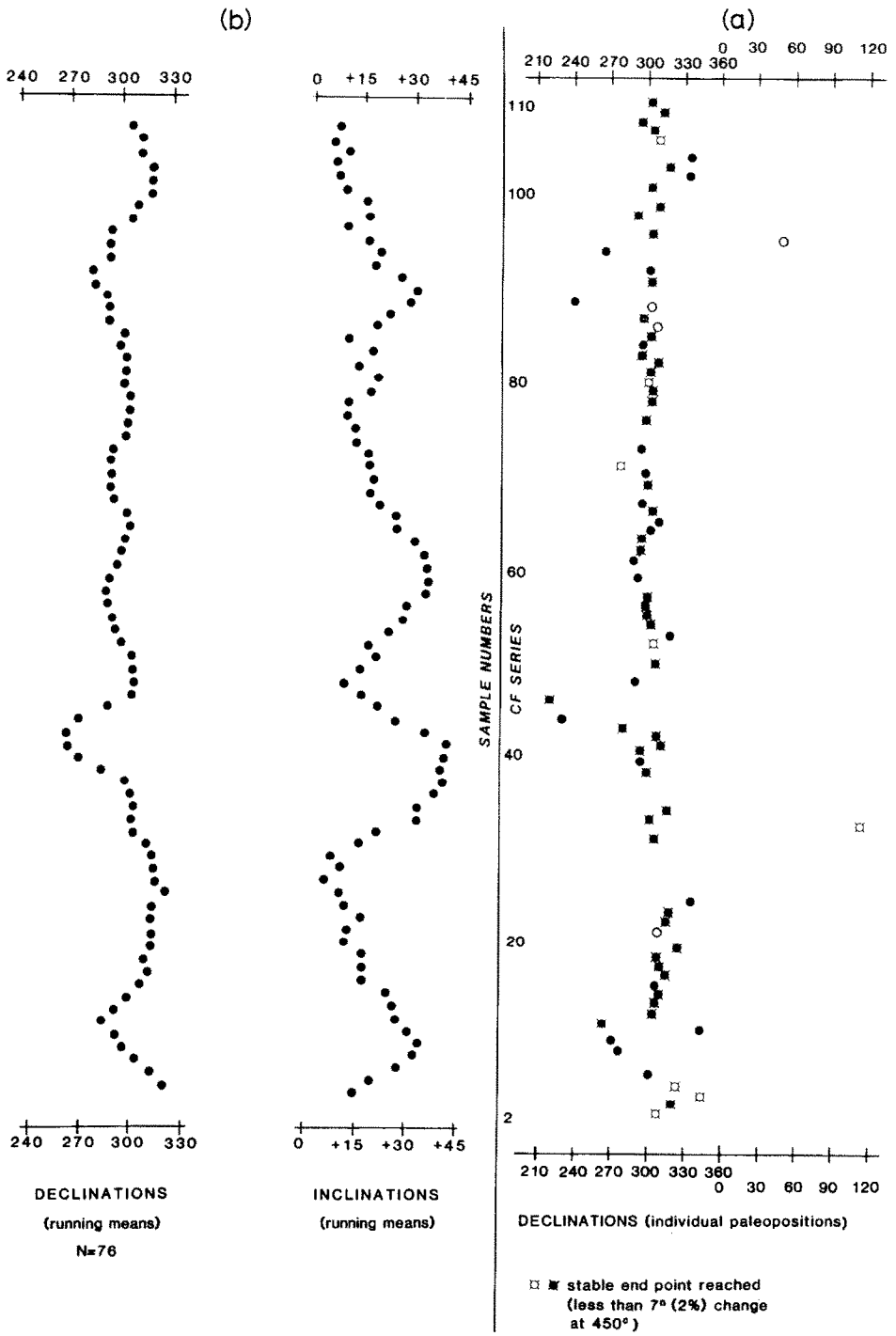


Fig. 11. A stratigraphic plot of declination and inclination values with respect to paleoposition after cleaning at 450°C. a. Individual declinations. b. After smoothing using a running mean of 5 samples and omitting anomalous results.

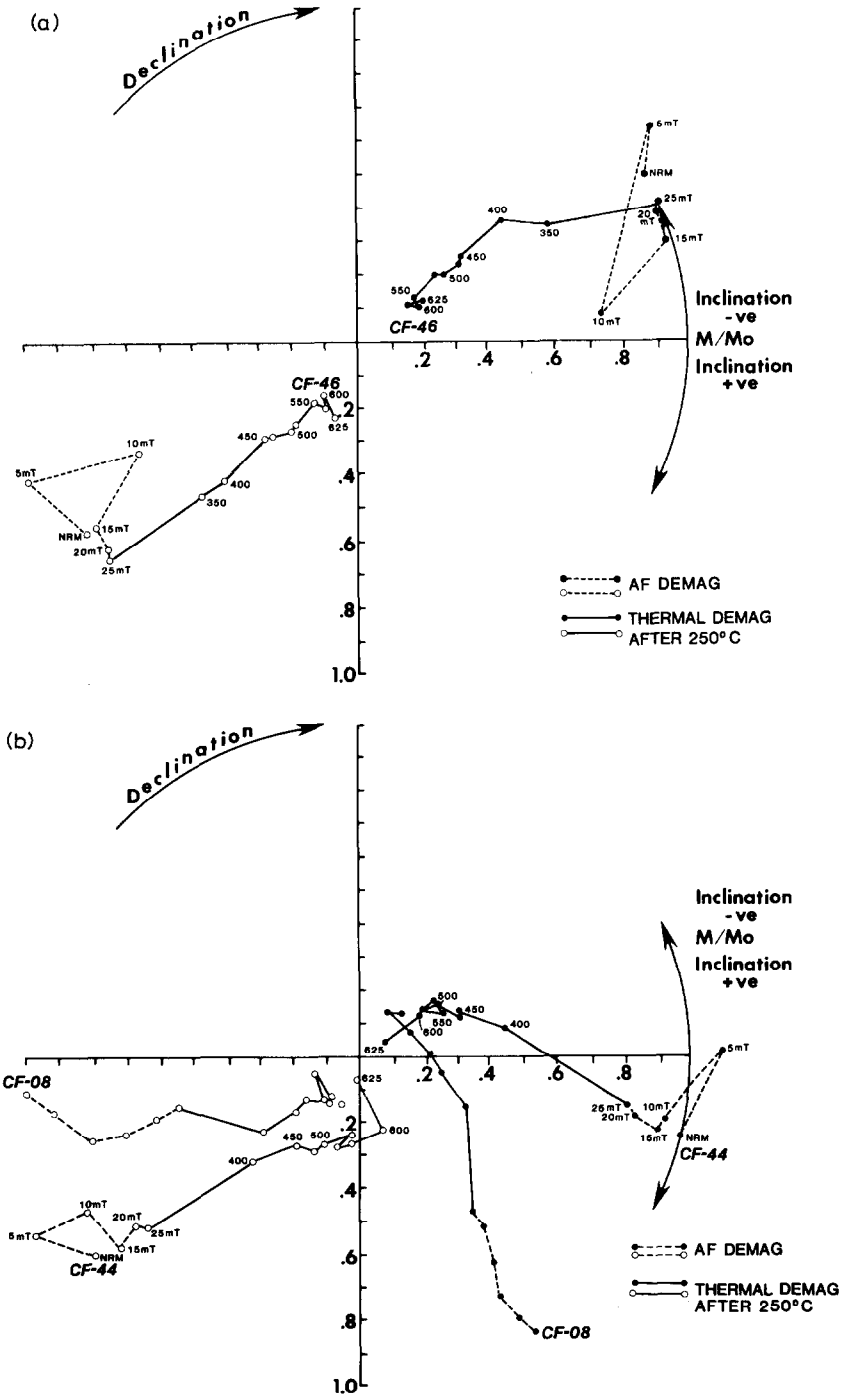


Fig. 12. Stable endpoint curves where no characteristic magnetization has been isolated up to 625°C. Vector subtraction indicates the removal of a component with a southerly declination and low negative dip when measured with respect to the field position.

DISCUSSION

In the few cases where superposed magnetic components were identified and separated, the harder component, when calculated with respect to the field position, approaches that of the Permian of the south of France (Kruseman, 1962; Zijderveld, 1975).

The great circle through the streaked but stable directions of magnetization, measured with respect to their field position, passes close to the Permian direction (Fig. 13). It therefore seems reasonable to interpret the magnetization as secondary, acquired during and/or subsequent to tectonic emplacement. As at no stage do we find evidence of normal directions, it would appear that the magnetization postdates the period of Carboniferous field reversals and was therefore acquired in post-Westphalian A times, probably within the Kiamian interval.

To account for the stable but streaked directions of the majority of the samples, it is suggested that the rocks acquired their magnetization at different times during emplacement of the Mt. Peyroux nappe. To account for the positive inclinations, the beds were rotated about a horizontal axis during emplacement and transport southward, as indicated in the cross sections shown in Fig. 1b.

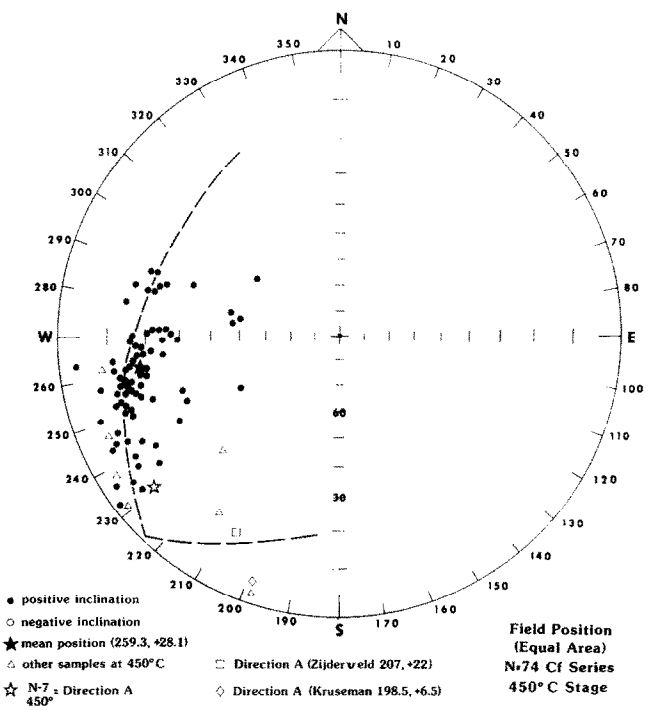


Fig. 13. Directions of magnetizations at 450°C, plotted with respect to field position. The Permian positions of Zijderveld (1975) and Kruseman (1962) are plotted.

As a good case can be made that the hematite of the griotte was formed at the time of deposition (see Jenkyns and Torrens, 1971), or at the latest during early diagenesis, this implies that the critical domain size may have been reached by crystal growth or recrystallization, presumably aided by thermal conditions in the tectonically transported rock mass. Since the magnetization is secondary, no contribution can be made with respect to any sedimentologic or magnetostratigraphic problem.

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